



DEPLOYMENT PAPER

PoC to Production: Why Most AI Deployments Stall

Engineering

Senior Leadership

Product Management

88% of AI proofs of concept never reach production. The technology works. The model is accurate. The demo impresses the board. Why does nothing ship? A root-cause analysis and a systematic remedy.

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The 88% Problem

95% of generative AI pilots fail to deliver rapid revenue acceleration (MIT NANDA 2025). 42% of companies abandoned most AI initiatives in 2025 (S&P Global). The pattern is consistent across every study: the AI works, the project stalls. The bottleneck is not the model — it is everything around the model.

The disconnect. Enterprise AI spending will reach \$407 billion in 2026 (IDC). 78% of organizations now use AI in at least one business function (McKinsey, 2025). Yet the conversion rate from pilot to production is lower than any major technology cycle in recent memory. Cloud computing conversion was ~60%. Mobile was ~70%. AI is at 5-12% depending on the study.

This is not a technology problem. The models work. The infrastructure is available. The talent exists. The failure is systemic — a combination of organizational dysfunction, measurement gaps, and architectural oversights that compound across every phase of the deployment journey.

The diagnosis. RAND Corporation's research on AI project outcomes (2024, based on interviews with 65 data scientists and engineers) found that more than 80% of AI projects fail — at roughly twice the rate of non-AI IT projects. The five root causes are consistently organizational: misunderstood problem definition, inadequate data, a technology-first mentality, insufficient infrastructure, and problem difficulty exceeding current AI capabilities. Gartner found that 63% of organizations lack AI-ready data. BCG found that only 36% of employees have been trained for AI transformation. Technology is not in the top three failure causes.



Sources: MIT NANDA State of AI in Business 2025 · RAND Corporation 2024 · S&P Global Voice of Enterprise 2025 · BCG AI Value Survey 2025

The remedy. Closing the PoC-to-production gap requires a systematic approach across four domains: data foundations, production architecture, organizational alignment, and continuous measurement. This paper presents a field-tested methodology for each domain, based on Govantis's work across telecom and critical infrastructure platforms.

The Four Gaps That Kill AI Deployments

Gap 1: Data Foundations

The most cited technical barrier to AI production deployment is data readiness. Gartner predicts that "through 2026, 60% of AI projects lacking AI-ready data will be abandoned" (Gartner AI Hype Cycle, 2025). This is not about data volume — most enterprises have more data than they need. It is about data quality, accessibility, and governance.

DATA PROBLEM	IMPACT ON AI	% OF ORGS AFFECTED
Data silos across departments	Models trained on partial data, poor generalization	63%
No production-grade data pipelines	Batch inference only, no real-time or streaming capability	57%
Data quality issues (missing, inconsistent, stale)	Model accuracy degrades in production vs. training	52%
Insufficient labelled data for fine-tuning	Teams default to zero-shot RAG — works in demo, fails in production	48%
Data privacy and compliance constraints	Production data cannot be used the way POC data was used	44%

Common pattern: A team builds a POC with a carefully curated, static dataset. The model performs at 92% accuracy. The demo impresses leadership. Then the team tries to deploy to production — and discovers that production data is messy, unlabelled, siloed across three databases, and subject to GDPR restrictions that the POC data was not. The model's accuracy drops to 67%. The project enters a "data remediation" phase that lasts 6-12 months. The sponsor changes jobs. The project dies.

Gap 2: Production Architecture

The POC runs on a laptop, a Jupyter notebook, or a single GPU instance. Production requires: automated retraining pipelines, CI/CD for model deployment, monitoring infrastructure, versioned model registries, A/B testing frameworks, and cost management. Few AI teams budget for this transition.

The architecture gap manifests in five predictable failure modes:

No Retraining Pipeline

The model is trained once on historical data. In production, data distribution shifts. The model degrades silently. Nobody notices until the business impact is material.

No Monitoring

Infrastructure monitoring (CPU, memory, latency) is confused with model monitoring (drift, accuracy, bias). The server is healthy. The model is failing.

No Rollback

When a model update causes regression, there is no mechanism to revert to the previous version. The team either fixes forward under pressure or takes the model offline entirely.

No CI/CD for Models

Software has CI/CD. Models are deployed manually. A developer copies a file to a server, runs a script, and hopes. Every deployment is a production incident waiting to happen.

Cost Estimation Gap

The POC runs on \$50 worth of cloud compute. Production at scale costs \$50,000/month in GPU instances — budget that was never requested because nobody calculated inference cost per request.

Gap 3: Organizational Alignment

RAND's finding — 84% of AI project failures attributable to leadership and strategy — is the most important statistic in this paper. The pattern is consistent across every study we reviewed:

ORGANIZATIONAL FAILURE MODE	EVIDENCE
No shared definition of success	RAND research (2024) identified "misunderstood problem definition" as the #1 root cause of AI project failure — the most fundamental failure mode, occurring before any technology is selected. Teams build what they think is asked. Leadership approves what they think is proposed. Neither validates.
Executive-employee misalignment	76% of executives feel positive about AI; only 31% of employees agree (BCG/HBR, 2025). Leadership pushes adoption. Teams resist what they do not understand.
No trained workforce	Only 36% of employees have been trained for AI transformation (BCG, 2026). The remaining 64% cannot effectively use or support the AI tools being deployed.
Vague or absent financial KPIs	Only 5% of companies achieve substantial AI value because only 5% define what "value" means before starting (BCG, 2025).
Ownership vacuum	CTO thinks AI is a data science problem. Data science thinks it is an infrastructure problem. The board thinks buying Copilot licenses is an AI strategy. Nothing ships.

Gap 4: Measurement Blindness

Organizations are spending aggressively — global generative AI investment tripled to roughly \$37 billion in 2025 — but most cannot answer a simple question: **What is the return on our AI investment?** McKinsey research (2025) found that over 80% of organizations report no meaningful enterprise-wide EBIT impact from AI despite widespread adoption. BCG corroborates this: only 5% achieve substantial AI value at scale. The remaining 95% are spending without measuring what matters.

The measurement gap: Without defined success metrics, every AI project is a success (the demo works) and a failure (nothing ships to production). The organizations that consistently ship to production — estimated at 5-10% of enterprises — are those that define their success metrics before the POC begins, not after it fails to scale.

Closing the PoC-to-Production Gap

Based on our experience across telecom and critical infrastructure platforms, closing the gap requires a systematic approach across four dimensions, reinforced by a shift in resource allocation.

The 10/20/70 Allocation Model

The most successful AI deployments follow a resource allocation pattern that is the inverse of what most organizations do:

DOMAIN	TYPICAL ORG ALLOCATION	SUCCESSFUL ORG ALLOCATION
Algorithms & Models	70%	10%
Technology & Infrastructure	20%	20%
People, Process & Change	10%	70%

Source: BCG, "Scaling AI Requires New Processes, Not Just New Tools", January 2026.

Most organizations spend 70% of their AI budget on model development and tuning — the part that already works. They spend 10% on the organizational change management that determines whether the deployment succeeds. Successful organizations invert this: they invest in data pipelines, production infrastructure, workforce training, workflow redesign, and change management. The model itself is a commodity.

Domain 1: Data Foundations

Data Pipeline Audit

Map every data source, pipeline, and transformation that the POC uses. Compare to production data availability. Identify gaps. This audit takes 1-2 weeks and prevents 6-month remediation cycles.

Continuous Data Governance

Implement data quality monitoring across all production data sources — completeness, consistency, timeliness. Link governance to model performance alerts.

Synthetic Data Strategy

For regulated domains where production data cannot be used freely, build synthetic data generation pipelines that preserve the statistical properties of real data.

Domain 2: Production Architecture

MLOps Baseline

Before any model reaches production, deploy: model registry, automated retraining pipeline (triggered by drift detection), A/B testing framework, and cost-per-inference dashboard.

CI/CD for Models

Treat model deployments the same as software deployments — automated testing, staged rollouts, rollback capability, versioning, and approval gates. No manual deploys.

Observability Stack

Four monitoring layers: infrastructure (CPU, memory, latency), model (drift, accuracy, bias), business (user engagement, conversion, retention), and cost (inference cost, infrastructure cost).

Domain 3: Organizational Alignment

Shared Success Definition

Before the POC begins, define: what does success look like? Who decides? What metrics will be used? At what threshold does the project go/no-go? Document it. Sign it.

Executive Sponsorship Contract

Identify the executive sponsor who owns the outcome. Not just funding — ownership. Monthly reviews. Escalation path. No delegation to a "senior stakeholder."

Change Management Plan

Before deployment, map: who needs to change their workflow? What training do they need? What resistance is expected? How will adoption be measured? Treat change management as a project deliverable, not an afterthought.

Domain 4: Continuous Measurement

AI Value Scorecard

Per model: inference cost, user adoption rate, accuracy/drift score, business impact (revenue, cost, satisfaction). Published weekly to the executive sponsor. No surprises.

Pilot-to-Production Pipeline

Track every POC through stages: ideation → validation → production → scaling. Measure stage transition rate and time-in-stage. If a POC has been in validation for more than 8 weeks, escalate. The goal: identify bottlenecks, not kill projects.

Cost Transparency

Every model has a cost-per-inference dashboard that includes: compute, data retrieval, API calls, human review (if applicable). Cost awareness drives architectural decisions — and prevents deployment-time budget surprises.

From POC to Production: A 12-Week Programme

The Govantis PoC-to-Production programme is designed to convert an existing AI proof of concept into a production-grade deployment in 12 weeks. The programme is structured in three phases, each with clear deliverables and decision gates.

01

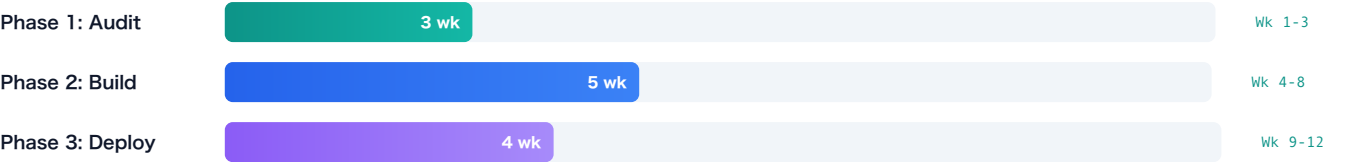
Phase 1: Audit & Baseline (Weeks 1-3)
Data pipeline audit, production architecture assessment, stakeholder alignment workshop, success metrics definition, cost estimation. **Deliverables:** production readiness report, architecture gap analysis, success criteria document, budget estimate.

02

Phase 2: Build & Integrate (Weeks 4-8)
Data pipeline remediation, MLOps infrastructure deployment, CI/CD for models, monitoring stack implementation, model registry setup. **Deliverables:** production infrastructure running, model re-trained on production data, monitoring dashboards live.

03

Phase 3: Deploy & Validate (Weeks 9-12)
Staged rollout (5% → 25% → 50% → 100% of users), A/B testing against baseline, model drift monitoring configuration, change management execution, success metrics tracking. **Deliverables:** model in production at full scale, deployment report, validated ROI, handover to operations team.



Success Metrics by Stage

STAGE	GATING CRITERIA	TARGET
Phase 1 → 2	Data pipeline validated, success metrics defined, executive sponsor confirmed	All three met
Phase 2 → 3	MLOps infrastructure live, model accuracy on production data ≥ POC accuracy, monitoring deployed	All three met
Phase 3 Complete	Model at full scale for 2+ weeks, drift < 5%, user adoption > 60%, ROI positive	≥ 4 of 5 met

What the Successful 5% Do Differently

Across every study and our own field experience, the organizations that consistently ship AI to production share five disciplines:

01

They define success before the POC
Before any code is written, they define: what is the metric that determines success? What is the threshold? Who decides? This prevents the "impressive demo, nothing ships" pattern.

02

They invest in data first, models second
Data pipelines are treated as production infrastructure from day one. The model can be replaced. The data cannot.

03

They treat AI deployment as a business transformation, not an IT project
Workflow redesign, change management, and workforce training are not afterthoughts — they are core deliverables.

04

They measure everything from day one
Cost per inference, model drift latency, user adoption rate, and business impact — tracked weekly, reviewed monthly.

The gap between AI ambition and AI execution is the defining challenge for technology leaders in 2026. The technology works. The organizations do not. Closing this gap is not a technical problem — it is a leadership problem with a technical solution.

Immediate Next Steps

01

Run a PoC-to-Production Audit

Two-week assessment of your current AI pipeline. Audit every POC against production readiness criteria. Classify each as: ready to ship (with gap list), stalled (identify blocker), or dead (cancel explicitly). **Estimativa Govantis — 2 weeks.**

02

Implement the Production Readiness Checklist

Define the minimum criteria that a POC must meet before it can enter the deployment pipeline. Non-negotiable: data pipeline validated, success metrics defined, executive sponsor assigned, MLOps infrastructure budgeted.

03

Shift to 10/20/70 Resource Allocation

Reallocate AI investment from model development (where 70% of budget goes today) to data infrastructure, production architecture, and change management. The model is not the bottleneck.

04

Build the Deployment Playbook

Standardise the process for moving an AI POC to production. Not a template — a practised, tested, improved process that every team follows. Iterate after every deployment.

05

Publish the AI Value Scorecard

Make AI deployment metrics visible to leadership: number of POCs, conversion rate, average time-to-production, cost per deployed model, ROI per model. Transparency creates accountability.

The bottom line: The difference between organizations that ship AI and organizations that demo AI is not technical sophistication. It is operational discipline. The 5% that succeed have boring, reliable processes for data, deployment, measurement, and change. The 95% that stall have impressive demos. Choose which one you want to be.



About Govantis

Govantis is an AI-native platform consultancy founded by engineers who spent decades inside Deutsche Telekom, Vodafone, BT, and NOS. We design architectures, lead platform migrations, and build governance programmes for the world's most complex infrastructure platforms.

Three capabilities. Built for your platform.

Complex Delivery

End-to-end execution on multi-vendor delivery programmes. Broadband, TV, connectivity, and AI infrastructure projects.

AI Products & Scale

Ideate, prototype, and scale AI-native products for complex operational environments. From monitoring to edge intelligence.

Governance & Strategy

Independent governance for complex multi-stakeholder programmes. AI readiness assessment. Regulatory alignment.

RDk

prpl Foundation

Deutsche Telekom

Vodafone

BT

NOS

Stop demos. Start shipping.

If AI deployment discipline matters to your platform, it matters to us.

info@govantis.pt

Rua Castilho, n.º 39, 8.º E · 1250-068 Santo António, Lisboa · Portugal

www.govantis.pt · linkedin.com/company/govantis

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